

Title: Multi - Agent AI Framework for Automated Software Development

Abstract:

Software development involves several stages — requirement gathering, planning, coding, testing, and review — typically handled by different team members, demanding significant time and coordination. This project proposes a Multi-Agent AI Framework for Automated Software Development, where four cooperating LLM-based agents collaborate to automate the core stages of the Software Development Life Cycle (SDLC) in a structured, spec-driven pipeline.

The framework is orchestrated using LangGraph, which manages a cyclic, stateful graph of four agents — Planner, Coder, Tester, and Reviewer — rather than a fixed linear pipeline. The Planner turns a plain-English requirement into a structured, Pydantic-validated specification with explicit acceptance criteria, which a human reviewer approves before any code is generated. To balance performance against free-tier rate limits, the system uses a mixed-model setup: the Planner and Reviewer, where robust reasoning matters most, run on Groq's larger openai/gpt-oss-120b model, while the higher-volume Coder and Tester run on the smaller, faster openai/gpt-oss-20b — both served through Groq's free tier at no cost. Every agent's output is forced into a validated JSON shape rather than relied upon as free text, and generated code is executed for real inside a sandboxed subprocess, one acceptance criterion at a time, rather than a claimed pass/fail. A capped retry loop routes failed tests or rejected reviews back to the Coder — carrying the specific failure reason forward — so the pipeline can recover from real failures without running away.

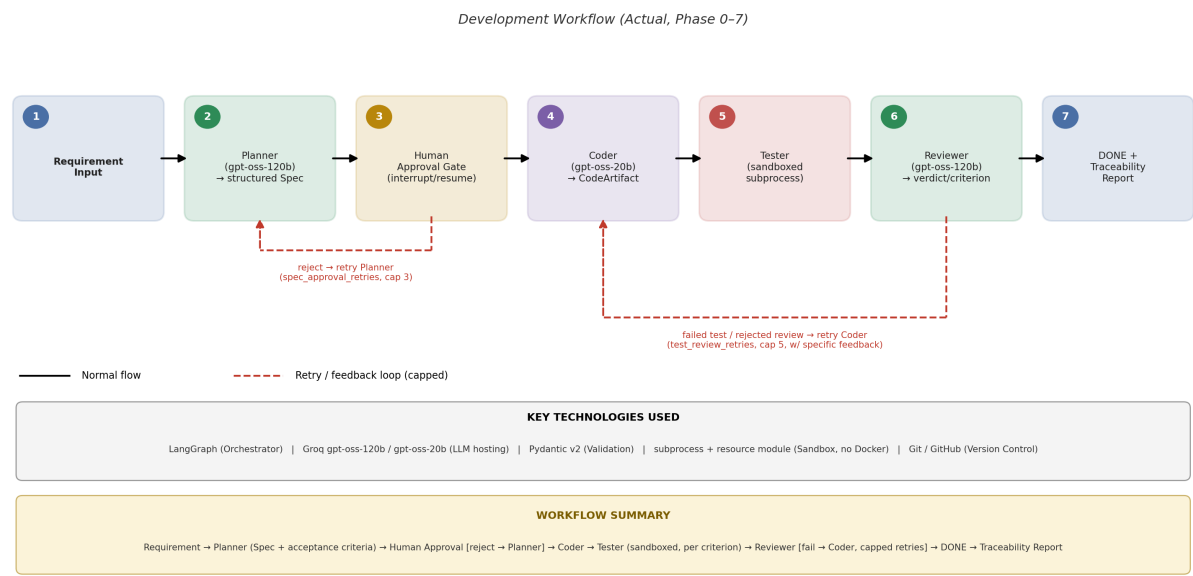
By interpreting natural language requirements and generating, testing, and reviewing software artifacts collaboratively, the agents work together to improve efficiency, reduce manual effort, and produce higher-quality, verifiably tested software. Every run concludes with an automatically generated traceability report tracing requirement, acceptance criterion, task, code, test result, and review verdict, readable without narration. The system helps developers by accelerating development cycles and catching errors early, while relying entirely on free-tier tools and APIs to keep the framework accessible.

Datasets:

HuggingFace	https://huggingface.co/datasets/BasitAliiii/sdlc-scenario-based-qa-dataset
Github	https://github.com/GPT-Laboratory/CodePori_v02
Kaggle	ai_multi_agent_interactions_2026.csv

Workflow:

Multi-Agent AI Framework for Automated Software Development



Project Members:

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